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MILITARY SPECIFICATION

* JACK, HYDRAULIC, TRIPOD MMU-129/E, 60 TON CAPACITY, 72- TO 144-INCH RANGE

1. SCOPE

1.1 This specification covers a 60-ton, hydraulic, tripod jack, designated MMU-129/E.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-P-5516	Packing, Preformed, Petroleum, Hydraulic Fluid Resistant, 160°F
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-C-6021	Casting, Classification and Inspection of
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Testing
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra-Low Temperature

FSC 1730

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

DRAWINGS

Air Force

43A13905	Connector Assembly - Hydraulic Hose, Female
56B6129	Socket - Ball Type Jack Pad

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

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3.2 Components. The jack shall consist of the following major components:

Item or System Name	See Requirement
Ram and cylinder assembly	3.7.1
Hydraulic pumping system	3.7.2
Supporting structure	3.7.3
Caster running gear	3.7.4.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. Materials used in the construction of the jack shall be protected against deterioration when exposed to climatic and environmental conditions likely to occur during service. This protection against deterioration shall be accomplished in a manner which will not prevent compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.2 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.3 Standard steels. Steels complying with ASTM or AISI standards shall be utilized in the design and fabrication of the jack and shall be corrosion resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered during storage or normal service.

3.4.3.1 Heat treatment. Heat treatment of steel materials shall be in accordance with MIL-H-6875.

3.4.4 Castings. Castings shall be designed and classified in accordance with MIL-C-6021.

3.4.5 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.5 Design and construction. The jack shall be designed as a tripod structure supporting an extendable, single-stage, hydraulic ram mounted in a vertical position at the apex of the tripod. The jack shall be equipped with a self-contained, hand operated, hydraulic pumping system and shall also have provisions for connection to an external hydraulic pumping source. The tripod legs shall be equipped with retractable swivel casters and shall have provisions for towing. Configuration of the jack shall be as shown on figure 1.

3.5.1 Capacity. The jack shall be designed for a rated capacity of 60 tons and shall have a proof load capability of 90 tons.

3.5.2 Structural design. The jack ram and cylinder shall be designed to include a safety factor of 1.75 based on the yield strength of the material. The remainder of the jack supporting structure shall be designed for a safety factor of 2.0 based on the yield strength of the material.

3.5.3 The jack shall be designed and constructed to assure that all parts and components are so assembled as to be incapable of working loose during service. The jack shall be designed to withstand the structural loads, strains, vibrations, and other conditions incident to operation, shipping, and storage.

3.5.4 Component arrangement. All components shall be arranged for maximum ease of operation and maintenance consistent with the utility and efficiency required to conform to the requirements specified herein.

3.5.4.1 All parts, such as pumps, seals, packaging, valves, and controls, shall be new and shall have proper clearances and adjustments. They shall so work together that the jack will supply the rated capacity without unnecessary strain.

3.5.5 Reliability. The jack shall have a minimum mean-time-between-failures (MTBF) of not less than 120 cycles with a reliability of not less than 0.990 for a 1-cycle mission at 0.9 confidence factor (see 6.3).

3.5.6 Maintainability. The jack shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

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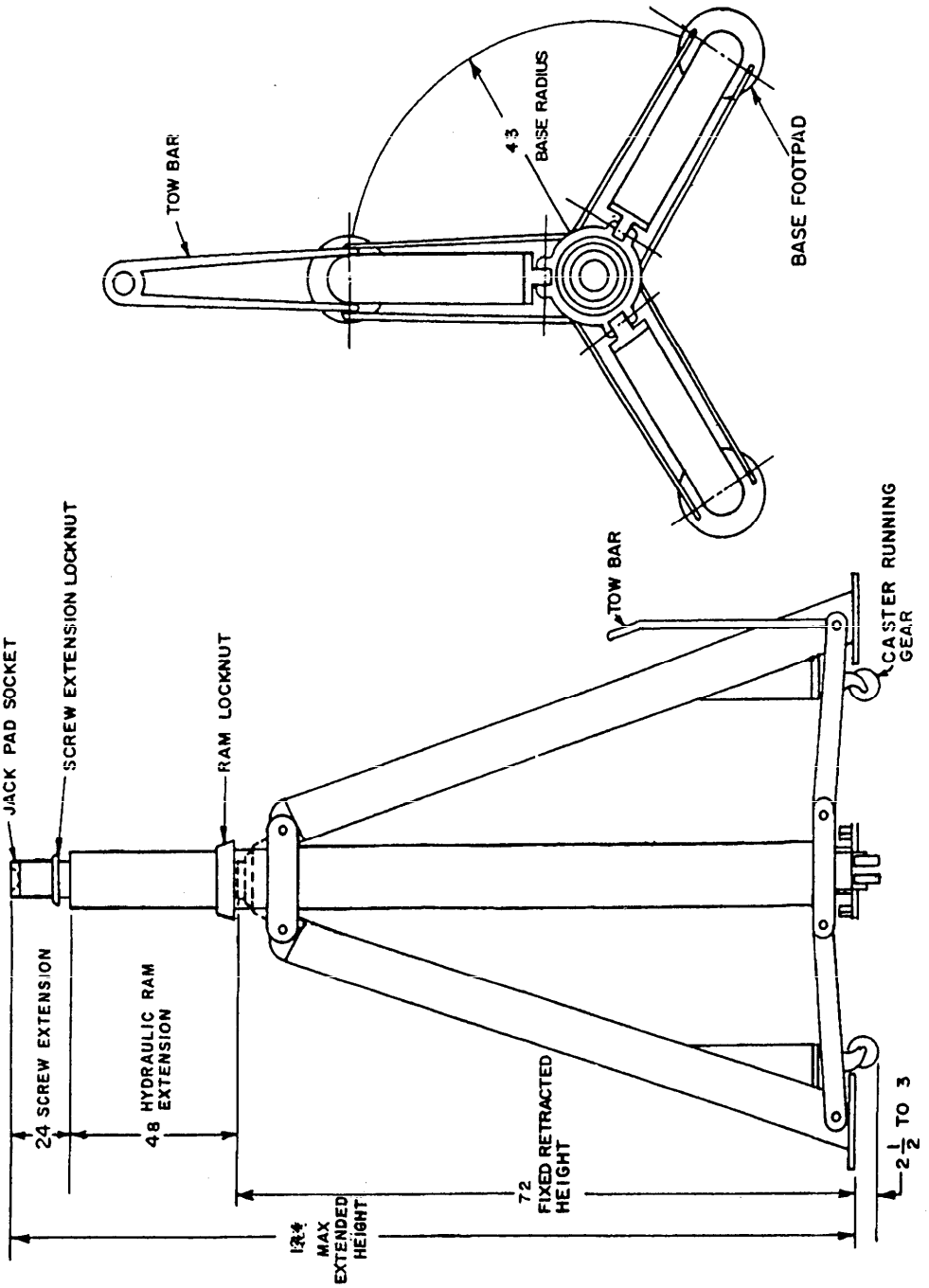


*FIGURE 1. Tripod Jack

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c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general purpose tools and equipment normally available commercially. Use of special purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum

f. Foolproofness - Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.7 Human engineering. The jack shall be so designed that human performance requirements under the range of environmental conditions indicated in 3.6.1 will be consistent with the characteristic of typical (5th to 95th percentile) maintenance personnel as described in MIL-STD-1472.

3.6 Performance

3.6.1 Environmental conditions. The jack shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -65° to +120°F

b. Storage temperatures ranging from -80° to +160°F

c. Exposure to salt fog

d. Dust as encountered in desert areas.

3.6.2 Lifting capability. The jack shall be capable of raising a load equal to 150 percent (90 tons) of the rated capacity of the jack without incurring failure or permanent deformation.

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3.6.3 Settling. When supporting a rated capacity load of 60 tons, the ram settling rate shall not exceed 0.020 inch per hour.

3.6.4 Hydrostatic pressure. The ram and cylinder assembly shall withstand an internal fluid pressure of 7,000 psi without leakage.

3.6.5 Side load capability. The jack shall withstand horizontal side loads, equal to 15 percent (9 tons) of the rated capacity load at the jack pad socket, with the ram and screw in the fully extended position.

3.6.6 Graduated side load. The jack shall withstand lifting the rated capacity load while a horizontal load, graduated between 0 and 15 percent, is applied to the jack pad socket as the ram is being extended.

3.6.7 Ram collapse capability. The unloaded ram shall be returned to the retracted position from any extended position with not more than 50 pounds applied vertical force.

3.6.8 Mobility. The jack shall meet the type I, class 2 mobility requirements of MIL-M-8090.

3.7 Details of components

3.7.1 Ram and cylinder assembly. The ram and cylinder assembly shall be designed for a ram extension capability of 48 inches under the load conditions specified in 3.6.

3.7.1.1 Hydraulic fluid pressure. The ram and internal cylinder diameter shall be designed for a hydraulic fluid operating pressure of 4,000 $\pm$50 psi at the rated capacity load of 60 tons. A capped tee fitting shall be provided in the hydraulic system to provide a facility for measuring the fluid operating pressure.

3.7.1.2 Bearings. The ram and cylinder bearings shall be of sufficient size and contact area to insure maximum stability and life under all ram-screw extensions and vertical-horizontal load conditions specified herein.

3.7.1.3 Packings. Ram packings and seals shall be of the O-ring type conforming to class B of MIL-P-5516 with V-type teflon backup rings. Equivalent materials shall require approval by the procuring activity.

3.7.1.4 Ram. The hydraulic ram shall be designed with an external thread and safety locknut to prevent settling of a load or dropping a load in the event of hydraulic system failure. The design of the thread shall provide the maximum lead allowable and still assure positive holding under rated capacity load. The external surface of the ram shall be coated in accordance with MIL-C-13924.

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3.7.1.4.1 Locknut. The internal thread of the locknut shall mate with the ram thread. The locknut shall be designed for manual operation and shall be equipped with handles for ease of operation.

3.7.1.4.2 Screw extension. A threaded screw extension shall be provided in the upper end of the ram assembly. The screw shall be designed for manual operation and shall be capable of extending 24 inches. The screw shall be provided with a stop to limit the extension to 24 inches. The screw extension length and thread bearing area shall be consistent with all load and extension conditions specified herein.

3.7.1.4.3 Screw extension locknut. The screw extension shall be equipped with a manually operated safety locknut. The locknut shall be designed for manual operation and shall be equipped with handles for ease of operation.

3.7.1.4.4 Jack pad socket. The upper end of the screw extension shall be equipped with a jack pad socket conforming to Drawing 56B6129.

3.7.1.4.5 Ram gage. The ram shall be equipped with a gage device to provide visual indication of the extended length of the ram. The gage shall be calibrated in inches.

* 3.7.1.5 Cylinder. The hydraulic cylinder assembly shall include the ram assembly and hydraulic pumping system. The nominal inside diameter of the cylinder shall be 6.187 inches. The internal walls of the cylinder shall be chrome plated. The length of the cylinder assembly shall be between 90 and 95 percent of the 72-inch vertical height of the retracted jack. The lower end of the cylinder shall provide mounting points for the hydraulic pump(s) and lower horizontal structural braces. The hydraulic fluid reservoir shall be located adjacent to the lower end of the cylinder or elsewhere as approved by the procuring activity.

3.7.2 Hydraulic pumping system. The hydraulic pumping system shall be designed for operation with hydraulic fluid conforming to MIL-H-5606. The system shall consist of a double-acting two-speed pump(s), necessary bypass, release and check valves, reservoir tubing, and fittings that will provide for a self-contained pumping capability.

* 3.7.2.1 Pump(s). The pump(s) shall be hand operated and capable of adjustment to provide for pumping fluid volume control. The ratio between the jack ram and pump piston shall be designed to require a force not greater than 75 pounds at the end of the pump handle to raise a load equivalent to the rated capacity of the jack. The length of the pump handle shall not exceed 30 inches. The stroke of the pump handle shall be at a height that can be operated by a man standing on the ground. The packing shall be of the O-ring type conforming to class B

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of MIL-P-5516 with V-type teflon backup rings. A screen of approximately 50 mesh, 0.007-inch diameter brass wire shall be provided on the pump inlet port.

3.7.2.2 Bypass valve. The jack shall be equipped with a bypass valve adjusted to 105 to 110 percent of the fluid pressure that is required to raise a capacity load. The bypass valve shall be located between the pump check valve and jack ram to prevent overload and shall vent into the reservoir.

3.7.2.3 External pumping source connector. The hydraulic circuit shall have provisions for operating the jack from an external hydraulic pumping source. A female connector conforming to Drawing 43A13905 shall be provided on the jack for this purpose. The hydraulic circuit shall be designed to provide positive assurance that pressurized fluid from the external pumping source will not enter the jack fluid reservoir during external pumping operations. A method to automatically isolate the fluid reservoir from the pumping circuit when the external pumping source is connected to the jack shall be provided.

3.7.2.4 Reservoir. The reservoir shall have a fluid capacity sufficient to fully extend the jack ram plus a 10-percent minimum reserve. The reservoir shall be equipped with an air relief vent, fill cap, drain plug, and sight glass type fluid level indicator. The fluid level indicator shall be marked to indicate the acceptable range of fluid level when the ram is fully retracted.

3.7.3 Supporting structure. A rigid symmetrical tripod structure shall support the ram and cylinder assembly. The structure shall provide the complete jack with stability and safety under all combinations of load and extended height conditions. The support structure design shall incorporate the use of pinned joints at all interfaces between the ram/cylinder assemblies, legs, and bracing. The base radius of the jack legs shall be as shown on figure 1.

3.7.3.1 Legs. The jack legs shall be designed as primary axial load carrying members and shall be of one-piece, hollow, tubular construction. The upper end of each leg shall so attach to the upper end of the cylinder assembly as to transfer the total vertical load of the ram/cylinder assembly to the legs. The lower end of each leg shall be laterally stabilized by horizontal bracing between each leg and lower end of the ram/cylinder assembly. Braces also shall be used to interconnect adjacent pairs of legs.

3.7.3.2 Bracing. Structural braces shall be designed and positioned to provide lateral stability to the lower end of the ram/cylinder assembly but shall not support any vertical loads imposed on the ram/cylinder assembly. Positioning of these braces shall also be such as to minimize the formation of bending moments in the legs.

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3.7.3.3 Leg footpads. Each jack leg shall be equipped with a footpad to distribute loads and prevent concentrated ground contact loading in excess of 400 psi when the jack is subjected to a rated capacity load.

3.7.4 Caster running gear. Swivel caster assemblies shall be provided at the lower end of each supporting leg for positioning and mobility purposes. The pivot point of each caster shall be between the centerline of each leg assembly and the vertical centerline of the cylinder. Caster wheels shall be 8 to 10 inches in diameter and equipped with a polyurethane tread. The two caster assemblies opposite the leg with the tow bar shall have provisions to lockout the swivel capability for towing control purposes.

* 3.7.4.1 Suspension. The jack shall be provided with caster units capable of both automatic retraction under a constant load and manual retraction. A retraction mechanism employing a spring-loaded vertical cylinder shall not be used. In the unretracted position, the casters shall suspend the jack structure to allow a 2-1/2 to 3 inch footpad ground clearance.

* 3.7.4.1.1 Automatic retraction. An automatic retraction mechanism shall be capable of uniformly lowering the jack to the ground under a constant load of approximately 450 ±50 pounds per caster as load is applied to the jack.

3.7.4.1.2 Manual retraction. A manual retraction mechanism shall allow for individual retraction of each caster. The retraction mechanism shall be hand operated and shall not require the use of hand tools.

3.8 Accessory equipment

3.8.1 Cover. A weatherproof protective cover with a holddown device shall be provided to cover the upper part of the retracted ram and cylinder assembly when the jack is not in use.

3.8.2 Tow bar. One leg of the jack structure shall be equipped with a tow bar and lunette eye. Provisions shall be made to secure the tow bar in the upright position, parallel to the jack leg, when the tow bar is not in use.

3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.10 Screw threads. Screw threads shall be in accordance with Handbook H28.

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block have the inscription "DO NOT STAND ON END" stenciled in 1-inch black

3.11 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with adequate means for preventing loss of proper tightness and adjustment. When subject to removal or adjustment, such parts shall not be swaged, peened, staked, or otherwise permanently deformed.

3.12 Finishes and protective coatings. Cleaning, painting, plating, anodic films, and chemical treatment shall be in accordance with MIL-STD-808. The exterior surfaces shall be painted yellow, film designation DG.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special marking shall be included:

CAPACITY 60 TONS.

3.14 Operating instructions. Instructions for operation of the jack by the integral pumping system and an external pumping source shall be prepared in an abbreviated form on permanently etched or engraved metal or plastic placards. The placards shall be permanently mounted on the cylinder assembly. The instruction markings shall be in accordance with MIL-STD-1472.

3.14.1 Additional instruction. The following marking shall be on or adjacent to the tow bar:

MAXIMUM TOWING SPEED - 5 MPH.

3.15 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, marking of parts and assemblies, alignment of parts, and tightness of assembly screws and bolts.

3.15.1 Welding. Where welding or brazing is used in the fabrication of the jack, the process used shall be in accordance with highest commercial standards and latest applicable codes of the American Welding Society. Welding shall not be resorted to as a repair measure, except that small blow holes in castings may be filled when not in critical areas, provided the responsible inspector is fully informed of all pertinent details and approved the repair procedure prior to its accomplishment. All welded joints shall be free from cracks, oxide inclusion, and injurious porosity.

3.15.2 Dimension and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may effect the interchangeability, operation, or performance of the jack, they shall be held or limited accordingly.

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3.15.3 Screw assemblies. All assembly screws and bolts shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of 28 to 32 inches Hg at a temperature of 25° ±5°C and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the changes in instrument readings.

4.3.2 Vertical loads. Vertical loads shall be applied to the jack through the jack pad socket and shall be colinear with the longitudinal centerline of the ram/cylinder assembly with the jack resting on a level surface.

4.3.3 Horizontal loads. Horizontal loads shall be applied to the jack through the jack pad socket. The jack shall be so positioned for maximum stability that the line of action of a leg is directed opposite to the direction of the horizontal load.

4.3.4 Hydraulic fluid pumping source. The integral hydraulic pumping system shall be employed for all load lifting tests specified herein, except 4.6.5.3.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One jack shall be subjected to the preproduction tests specified in 4.4.3.

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4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of:

a. Individual tests

b. Sampling plan and test.

4.5.1 Individual tests. Each jack shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Hydrostatic pressure and leakage

c. Bypass valve test.

4.5.2 Sampling plan and test

4.5.2.1 Periodic sampling. Five jacks shall be selected at random from every 50 or fraction thereof produced and subjected to the staticproof test described in 4.6.4.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

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4.6 Test methods

4.6.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Initial inspection. The jack shall be disassembled to the maximum practical extent and completely inspected. Each component shall be checked as to weight, dimensions, finishes, et cetera. This data shall be recorded and compared with the data obtained in 4.10.

4.6.3 Operational tests. The jack shall be operated under the following conditions to insure that the performance of the various components comply with the requirements specified herein:

4.6.3.1 Hydrostatic pressure and leakage. The ram and cylinder assembly shall be tested by applying an internal hydraulic pressure of 7,000 psi to the assembly. This test shall be performed with an appropriate load or restraint on the ram to preclude damage to the ram-cylinder stops. There shall be no evidence of external leakage or deformation of assembly or component parts.

4.6.3.2 Settling. With the jack in the fully extended position and supporting a fully rated capacity load and the ram locknut placed in nonoperating position, the jack shall be allowed to stand for 1 hour. The height shall be measured at the beginning and at the end of this period. The rate of settling of the jack shall not exceed 0.020 inch per hour under rated capacity load, any excess of this distance shall be cause for rejection.

4.6.3.3 Bypass valve. Each pump shall be tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted as specified in 3.7.2.2.

4.6.3.4 Ram locknut. The ram locknut shall be tested with the jack ram at the fully extended position and supporting a rated capacity load. The hydraulic system shall be instantaneously vented to simulate a hydraulic failure. The locknut shall hold without dropping the load. Failure to hold the load shall be cause for rejection.

4.6.4 Static proof test. After the bypass relief valve has been adjusted for overload, the jack shall be tested with the screw extension in the fully extended position by raising a 90-ton load vertically from the lowest position of the hydraulic ram to the fully extended position. While in the extended position, the vertical load shall be reduced to 60 tons and maintained during application of a 9-ton horizontal load applied to the jack pad socket of the extended screw extension. To complete the cycle, the horizontal and vertical

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loads shall be removed and the jack ram allowed to retract to its original position with not more than 50 pounds applied to the top of the ram. There shall be no evidence of external leakage of hydraulic fluid from the ram or cylinder assembly. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5 Endurance tests. With the screw extension retracted, the jack shall be subjected to the following tests:

4.6.5.1 Vertical lift. The vertical lift test shall consist of 200 cycles of raising and lowering a vertical load of 60 tons. One cycle shall consist of raising the hydraulic ram under load from the fully retracted position to the fully extended position and return to the fully retracted position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5.2 Dynamic side load. The dynamic side load test shall consist of 60 cycles of raising a 60-ton load vertically with a graduated 9-ton horizontal load. Twenty cycles shall be performed with the jack positioned as described in 4.3.2 with each leg alternately positioned opposite to the horizontal load. The horizontal load on the jack pad socket shall be gradually increased from zero in the fully retracted position to 9 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5.3 External pumping. Ten cycles of lifting and lowering a capacity load of 60 tons shall be conducted with hydraulic fluid pressure furnished through the external pumping source connector. The positive protection provision specified in 3.7.2.3 shall be demonstrated. Failure of any part of the jack during this test shall be cause for rejection.

4.6.6 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810:

4.6.6.1 Low temperature. The jack shall be subjected to low temperature in accordance with method 502, procedure I. At the end of the exposure and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than a 50-pound force applied to the ram. For this test, the jack may be serviced with hydraulic fluid conforming to MIL-H-81019. The number of full strokes necessary to fully extend the ram shall be recorded at both $+70^{\circ}$ and -65°F. The full strokes required to fully extend the ram at -65°F shall be not more than double the strokes required at $+70^{\circ}$F.

4.6.6.2 High temperature. The jack shall be subjected to high temperature in accordance with method 501, procedure I. At the end of the exposure and

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with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than a 50-pound force applied to the ram. During this test, there shall be no evidence of external leakage of hydraulic fluid from the ram-cylinder assembly.

* 4.6.7 Mobility. The jack shall be towed over a level surface and an obstruction course to determine the general functioning of the running gear and the towing stability of the jack.

* 4.6.7.1 Level surface. The jack shall be towed over flight line hangar area type surfaces for a distance of 5 miles. The tests shall include 5 sudden stops and not less than 10 full turns to the right and 10 full turns to the left at the maximum towing speed of 5 mph. The jack shall be maintained under towing control at all times. At the completion of this test, the jack shall be inspected for excessive wear, distortion, or loosening of any assembly or component part.

* 4.6.7.2 Obstruction course. The jack shall be towed over an obstruction course for a distance of 2 miles at a speed of approximately 2-1/2 mph. The course shall consist of abrupt 1-inch high obstructions spaced at alternate 5- to 6-foot intervals perpendicular to the direction of towing.

4.6.8 Serviceability. The jack shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given to maintenance with a minimum number of tools, maintenance with general-purpose tools and equipment, servicing and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of contamination, residue, dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6.9 Final inspection. Upon completion of all tests, the jack shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. Measurements shall be made of all parts and components to determine wear and distortion, and shall be compared with the data recorded during the mechanical inspection (see 4.6.2). Wear, failure, or distortion that requires rework or replacement of parts affected shall be cause for considering the affected part as having failed to complete the tests satisfactorily. Jacks not conforming to this requirement shall be rejected.

4.6.10 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

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4.6.11 Maintainability evaluation. The jack shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning and drying. Each jack shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1, D-4, or D-5, as applicable, of MIL-P-116.

5.1.1.2 Unit protection. Each jack shall be preserved and packaged in accordance with MIL-P-116, method I. Exterior surfaces subject to corrosion shall be coated with type P-2 preservative. The hydraulic system shall be filled or flushed with fluid conforming to MIL-H-5606 or MIL-H-6083 and sealed. Each jack shall be disassembled where practicable and fully retracted. The towbar shall be secured in an upright position to the jack leg. Attaching parts accompanying the disassembled jack shall be preserved, bagged, or otherwise secured, appropriately identified, and attached to, or adjacent to, the attached fitting.

5.1.2 Level C. The jack shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The jack, preserved and packaged as specified in 5.1.1, shall be packed in a wood-cleated box with a load bearing base (skidded) conforming to MIL-B-26195, type II (overseas). The superstructure of the box shall be fabricated in accordance with PPP-B-601. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

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* 5.2.2 Level B. Level B packing shall be the same as level A, except the container shall conform to MIL-B-26195, type I (domestic).

* 5.2.3 Level C. The jacks shall be packed in a manner which affords adequate protection, at the lowest rate, against damage during direct shipment from the supply source to the first receiving activity. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice provided such meets the requirements of this level.

* 5.3 Physical protection. Interior cushioning, anchoring, bracing, and blocking shall be in accordance with MIL-STD-1186.

* 5.4 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Jack, Hydraulic, Tripod MMU-129/E, 60 Ton Capacity.

6. NOTES

6.1 Intended use. The MMU-129/E jack is intended for use on the C-5 aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction testing (see 4.4).

* c. Applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{MTBF}$$

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The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees of freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Cycle. One cycle consists of raising and lowering the hydraulic ram, under rated capacity load, from fully retracted position to fully extended position and returned to the fully retracted position.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 1730-F152

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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